Highlights

HomeFit: An Explainable Semantic Embedding Framework for Personality-to-Interior Design Mapping Using Large Language Models and XAI

Zainab Asif Jawed, Muhammad Rafi, Syed Zain-Ul-Hassan

- Introducing an explainable, personality-driven generative framework for cross-modal personality-to-design mapping, integrating textual personality inference with visual interior representations extracted from design scenes to generate semantically aligned and psychologically coherent design recommendations.
- Developing a retrieval-augmented generation pipeline using CLIP, BLIP, and SentenceTransformer embeddings to retrieve pre-encoded personality and interior representations for LLaMA-based context-aware design recommendation generation.
- Demonstrating that LoRA fine-tuning on clustered personality-style groupings improves semantic fidelity, aesthetic coherence, and cross-personality generalization, providing evidence of effective multimodal information fusion and adaptive style personalization.
- Providing multi-layered explainability through SHAP-based feature attribution and structured outputs, enabling step-wise interpretation of key words driving personality prediction and subsequent design recommendations to enhance user trust and satisfaction.

HomeFit: An Explainable Semantic Embedding Framework for Personality-to-Interior Design Mapping Using Large Language Models and XAI

Zainab Asif Jawed[a,*], Muhammad Rafi[b] and Syed Zain-Ul-Hassan[c]

[a]Department of Computer Science, FAST National University of Computer and Emerging Sciences, Karachi, Pakistan
[b]Department of Artificial Intelligence,FAST National University of Computer and Emerging Sciences, Karachi, Pakistan
[c]Department of Computer Science, FAST National University of Computer and Emerging Sciences, Karachi, Pakistan

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ABSTRACT

Recent improvements in artificial intelligence have made it possible for recommendation systems to personalise even more. In the automation of interior design, it is still hard to turn homeowners' psychological traits into clear and understandable aesthetic suggestions. Current methodologies frequently regard personality inference and design generation as discrete tasks, lacking a systematic framework to correlate psychological traits with visual design elements.To address this gap, we propose an individual, explainable generative framework based on a principled strategy for mapping personality to design.MBTI personality types are inferred from textual data using multiple embedding representations, with BERT yielding the most effective feature encoding. Classification using XGBoost achieves 85% accuracy, while a stacked ensemble further improves performance to 91%, demonstrating the effectiveness of model fusion for personality prediction. SHAP-based analysis identifies the linguistic features most influential in personality prediction, ensuring transparency in the inference process.The core contribution lies in a structured representation framework that associates inferred psychological profiles with interior design features, enabling context-aware recommendation generation through a retrieval-augmented architecture. The retrieval-augmented generation module produces initial recommendations, which are further refined through parameter-efficient fine-tuning using LoRA to improve semantic relevance and stylistic consistency.Extensive evaluation demonstrates robust generalization across personality types and a measurable reduction in the aesthetic–personality gap. The framework establishes a scalable human-centered paradigm for interior design automation by generating interpretable recommendations that link linguistic personality indicators with interior design features.The implementation and experimental code are publicly available for research reproducibility at the following repository: https://github.com/HomeFitCode

1. Introduction

The evolution of technology has consistently influenced architectural and interior design, with each innovation leaving a lasting mark on creative processes and design practices [1]. Recently, rapid developments in artificial intelligence, data science, and human-computer interaction have introduced tools such as generative models, retrieval-augmented systems, and explainable AI [2]. These advancements have accelerated design workflows and enhanced designers' ability to communicate and translate conceptual intent into realizable outcomes [3]. Decision-making, personalization, and interpretation are some of the direct contributions of the modern AI systems, compared to the previous digital tools, which were primarily auxiliary. This represents a significant paradigm shift in design style.It is a substantial challenge to align the psychological identity of a homeowner with the stylistic result offered by designers despite the growth in interior design. Homeowners frequently struggle to articulate explicit aesthetic preferences and they tend to give biased likes and dislikes. Consequently, designers have to make out vague descriptions or rely on intuition or perform iterative manual refinement. It is not only computationally and economically inefficient but also costly, often leading to increased burden on both stakeholders. Although scientific studies in the fields of psychology and design have always emphasized the correlation between personality traits and aesthetic preferences, these findings have remained mostly qualitative, and have never been put into practical and computational models [4]. Moreover, existing AI-based design systems are often treated as black boxes, generating layouts or styles without providing explicit rationale for their suitability to individual clients. This lack of transparency reduces the degree of trust and adoption, and designers do not have evidence-based tools to support their proposals, and homeowners do not trust the findings. In response to these limitations, the present study introduces a novel framework that integrates explainability, personality prediction, and AI-driven recommendations into a unified system. Homeowners' preferences, captured through simple like/dislike interactions, are translated into personality characteristic predictors, accompanied by clear word-level explanations to ensure transparency and confidence. Leveraging large language models enhanced via retrieval-augmented generation and LoRA fine-tuning, the framework delivers interpretable,

*Corresponding author
zainabasifjawed@gmail.com (Z.A. Jawed); muhammad.rafi@nu.edu.pk (M. Rafi); zain.hassan@nu.edu.pk (S. Zain-Ul-Hassan)
ORCID(s):
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context-sensitive design guidance. Unlike traditional methods, which rely on designers' intuition, expensive consultations, or opaque algorithms, this system allows homeowners to understand precisely which design elements align with their personalities and provides designers with evidence-based justifications, reducing revisions and eliminating intermediaries. Ultimately, the framework offers a scalable and interpretable methodology that bridges psychological theory and real-world design practice, benefiting both technical and non-technical audience in the interior design domain.

1.1. Problem Statement

The homeowners, as well as the designers, are affected by the major drawbacks of the existing systems of suggesting home interior designs. Since no known correlations between psychological qualities and design styles exist, they are based on demographics or superficial preferences rather than on structured and personality-driven understanding. Moreover, most AI-based models are functioning as black box and do not provide explanations or transparency, removing confidence of the homeowners, preventing designers to justify their actions, and leading to costly, repetitive adjustments. Moreover, such systems are often not contextually optimized as they do not apply higher-order learning interventions and fine-grained cognitive cues, and their outputs tend to look and feel beautiful, yet are psychologically invalid.

These inefficiencies undermine the effectiveness, personalisation and reliability of AI-assisted design. They require an explainable system that is personality-based and predicts user characteristics, an authenticated system that does it openly and generates context-driven, optimized recommendations. This would provide the homeowners with the idea why certain styles would fit their personalities and provide the designers with evidence-based reasons to minimize the revisions, decrease the costs, and provide more meaningful outputs.

1.2. Research Contributions

This study makes the following contributions to the field of AI-driven, personality-aware interior design:

1. **A Novel Personality-Aware Design Framework:** We propose an end-to-end framework that leverages personality prediction to generate personalized interior design recommendations. By aligning homeowners' psychological traits with design aesthetics, the system bridges the gap between psychological theory and practical design, enabling adaptive, context-aware, and meaningful design solutions.

2. **Explainable and Transparent Recommendations:** The framework integrates explainable AI techniques that highlight the key linguistic and personality factors influencing predictions. This provides homeowners with clear justifications for design suggestions and equips designers with evidence-based reasoning, thereby enhancing user trust, adoption, and interpretability of AI-assisted design systems.

3. **Retrieval-Augmented Generation with LoRA-Based Adaptation:** A retrieval-augmented generation (RAG) module using LLaMA produces initial recommendations, which are used for parameter-efficient LoRA fine-tuning to enhance generation quality. The framework is evaluated across multiple personality types using semantic embedding-based metrics, showing improved semantic relevance, personality consistency, and generalization.

Collectively, these contributions establish a scalable, interpretable, and personality-driven methodology for interior design, providing a benchmark for the integration of psychological modeling, AI-based optimization, and user-centered decision support in real-world applications.

2. Related Work

The study of personality-informed design personalization has gained significant attention across psychology, design studies, and artificial intelligence. While existing research consistently demonstrates that personality traits influence aesthetic preferences, translating these insights into transparent, data-driven, and computationally scalable systems remains a key challenge.This section reviews prior work across four interrelated areas: relationships between personality traits and design preferences, recommendation systems in aesthetic domains, dataset limitations in personality–design mapping, and emerging approaches to explainable and privacy-aware personalization. It further identifies existing research gaps that motivate the proposed framework.

2.1. Personality Traits and Interior Design Preferences

Environmental psychology and design studies have long focused on the impact of personality on aesthetic decisions. The Big Five, the Enneagram, and the Myers-Briggs Type Indicator (MBTI) are examples of personality theories that have been widely used to explain differences in interior preferences. For example, MBTI-based studies show systematic stylistic differences: introverts (e.g., INFJ, ISFJ, INTJ) prefer minimalist, natural, and emotionally resonant spaces, while extroverts (e.g., ESTJ, ENFP, ENFJ) prefer functional, modern, and high-tech interiors [3], [1]. There are also subtle differences between judging and perceiving types, which correspond with organized vs eclectic design orientations, and between sensing and intuitive types, such as pragmatism versus abstraction. These connections are supported by the Big Five framework. Extraversion is linked to vibrant and socially acceptable designs, conscientiousness to well-organized and practical layouts, and high openness to eclectic and artistic spaces [4], [5]. The Enneagram deepens motivation. For instance, Type 4 personalities (The Individualist) value design as a means of self-expression, but Type 6 personalities (The Loyalist) prefer safe and peaceful environments [6]. Moreover, there are trait-related outcomes, neuroticism is often associated with regulated and contained design, and extraversion is related to the preferences to more spacious and light spaces. Such tendencies are also developed on the basis of demographic and cultural factors [6] [7]. The design preferences have been found to differ greatly across age groups, genders, and cultural backgrounds, which is why research has also provided a systematic approach to the problem with a model of four simultaneous aspects that connect environmental stimuli such as shape, material, and light to personality traits , contextual Situation and cognitive-affective reactions factors. This body of work highlights the significance of personality in aesthetic judgment while revealing a gap in its translation into predictive computational systems.

2.2. Recommendation Systems in Aesthetic Domains

Recommendation systems have progressed a great deal regarding the lifestyle and aesthetic industry, more specifically the domain of fashion, travel [8], and also music [9].These systems are normally based on behavioral cues, which usually include the browser history, clicks, ratings, and personal data. Although these indicators are helpful in identifying preferences, they can lack psychological support and raise privacy concerns, particularly when user data is limited during cold starts. Computational techniques have been used in design and architecture with encouraging but incomplete outcomes. To match material features in furniture and door design, random forest classifiers have been used [10]. High accuracy in distinguishing interior design types from visual inputs has been attained by deep learning models like VGG16 [6]. With draw-to-shade methods for daylight control [10] and augmented reality systems including AI-generated design options [11], generative AI has increased personalization possibilities. Despite these advancements, the majority of systems still operate as non-transparent black-box models, producing outcomes that are difficult to interpret. Although personality-driven recommendation has been tested in related domains like media [9] and travel [8], these systems' scalability is limited by their reliance on long questionnaires and static profiles. While some research started connecting personality to design ([5], [1], [12]), they did not develop scalable, explainable recommendation pipelines. Consequently, the literature continues to be rich in conceptual linkages but deficient in design recommendation systems that are psychologically interpretable.

2.3. Dataset Limitations and the Challenge of Mapping Personality to Design

The lack of connected datasets continues to be a barrier to the advancement of personality-informed design suggestion. Though they lack psychological dimensions, large-scale visual repositories like the Room Interior Dataset, InteriorNet [13], and MIT Places [14] provide detailed annotations of spatial and stylistic aspects. On the other hand, personality databases like ZKA-PQ, Big Five, Enneagram, and MBTI offer trustworthy psychological evaluations but lack aesthetic or design elements. Because of this gap, scholars have created heuristic mappings. For example, some studies mapped Enneagram types to furniture and lighting choices ([5], [10]); others linked the Big Five dimensions to spatial and functional preferences [15]; still others matched MBTI types with specific design features like layout and lighting [1]. These mappings have provided directional insights, but they are nonetheless coarse, static, and context-dependent because they are frequently based on surveys or small-scale investigations. Attempts to tabulate mappings, such as linking ENTJ to modern or bohemian styles and INTP to industrial or vintage design, have not yet resulted in computer pipelines that can learn from massive amounts of data. A dataset or methodological framework that methodically integrates personality traits and design aspects into a machine-readable foundation is currently lacking in the field. The scalability, robustness, and practical implementation of computational models for personality-based design personalization have been hampered by this lack.

2.4. Toward Explainable and Privacy-Conscious Personalization

Explainability and privacy have become essential needs as AI applications expand into areas including private space and personal taste. Users find it difficult to grasp the mechanics underlying recommendations or to trust them in the absence of transparency [2] [16]. In the meanwhile, depending too much on behavioral logs presents ethical and privacy issues [17], particularly when used in private settings like home interiors [7]. Recent developments show only partial solutions to these problems. High-accuracy style detection is made possible by convolutional neural network classifiers [6], user-guided personalization is made possible by generative approaches, and immersive exploration of design options is made possible by augmented reality environments [17]. Nevertheless, these techniques seldom incorporate psychological frameworks or yield comprehensible, word-level descriptions of their results. Large language models now offer a way to overcome these drawbacks by integrating knowledge integration, computational efficiency, and natural language explanations into a single framework, especially when improved with retrieval-augmented generation and refined using low-rank adaptation.

3. Synthesis of Research Gaps and Present Study

There are four key research gaps identified across these streams. First, although it has been demonstrated that personality influences design preferences, research has not moved from descriptive to predictive, explicable computational models. Second, despite technological advancements, recommendation systems in design remain largely non-transparent and lack deep psychological modeling of user preferences. Third, the integration of personality traits and aesthetic attributes into a unified, machine-readable representation has been limited by dataset availability and structural constraints. Fourth, the current systems need a framework that integrates explainability, privacy-conscious personalization, and psychological validity. The current work takes a thorough and innovative approach to addressing these issues.Simple like or dislike inputs are used to predict personality traits, which reduce user load and prevent intrusive data collection. Word-based explainability is used to validate these predictions, enabling homeowners to comprehend and have faith in the model's logic. A coherent computational learning framework is established by systematically aligning two heterogeneous datasets—one psychological and one aesthetic—based on prior conceptual mappings.Lastly, personalized design recommendations that are precise, transparent, and privacy-conscious are produced by an artificial intelligence pipeline using big language models that have been improved using retrieval-augmented generation and refined with low-rank adaption. This study presents an end-to-end framework for explainable, personality-aware interior design customization by integrating machine learning, large language models, and established principles of personality psychology.

4. Methodology

This paper introduces a multi-phase, modular process of generating explainable, privacy aware, and psychologically consistent interior design suggestions. The proposed framework represented in Figure 1, which is the main roadmap of the methodology as it illustrates how each step turns user input into contextually relevant, psychologically consistent, and interpretable design output. As shown in Stage 1, the process starts with MBTI (Myers-Briggs Type Indicator) prediction of personality types, along with explanatory ability, which forms the psychological basis upon which further recommendations will be made. Stage 2 maps the predicted personality type to extracted interior design attributes, and the relationship between each individual trait and applicable aesthetic attribute is made to produce a structured personality-to-style dataset, which serves to guide the recommendation generation. The Stage 3 is based on the Retrieval-Augmented Generation (RAG) architecture where relevant design information is retrived considering the context of a particular design and the Stage 4 generates the initial interior design recommendations with the Large Language Model (Llama) and then refined with Low-Rank Adaptation (LoRA) to achieve optimal, personality-aligned design solutions that do not violate design constraints and user preferences. Lastly, Stage 5 adds an explainability module that gives clarity to the process of personality prediction and to the suggestion itself by showing the linguistic properties of user input that affect personality classification and which the essential words in the generated recommendations, thus making the homeowners realize how the given personality characteristics directly translate into the design outputs that the system recommends. All five steps represented in Figure 1, together, are comprehensively incorporated in this approach in order to generate individualized, cognitive-based, and interpretable interior design optimal.

4.1. Personality Prediction and Explainability

The initial stage, highlighted in Figure 1, focuses on analyzing textual data to predict the user's personality. This step is critical, as the accurate identification of personality types forms the foundation for mapping user preferences to appropriate interior design styles. The MBTI dataset, comprising 16 personality types derived from four binary dimensions—Introversion/Extraversion, Sensing/Intuition, Thinking/Feeling, and Judging/Perceiving—was adopted due to its structured categorical representation and extensive application in personality-aware computational systems. To get the dataset ready for modeling, data preparation was done. Tokenization, stopword and special character removal, and regular text cleaning procedures were all part of this. Despite being clean, the dataset revealed a natural class disparity among the 16 personality types. In order to establish homogeneity and enable the models to learn equitably across all personality categories, Random Over Sampling (ROS) was used. Three embedding methods were assessed in order to transform the cleaned text into numerical form. GloVe, BERT, and TF-IDF. GloVe added word co-occurrence for semantic relevance, whereas TF-IDF offered a baseline representation based on word frequency. However, the most accurate and semantically rich representations were generated by BERT embeddings due to their deep contextual learning. BERT was therefore selected for every downstream duty. These embeddings were used to train six machine learning models: Random Forest, Logistic Regression, Linear SVC, XGBoost, SGD, and CatBoost. With 88% accuracy, the combination of BERT and XGBoost produced the best standalone performance. A stacking ensemble strategy that used Random Forest, SVC, and XGBoost yielded further gains, increasing prediction accuracy to 91%. For downstream recommendations to be pertinent to users' true personalities, this accuracy was essential.

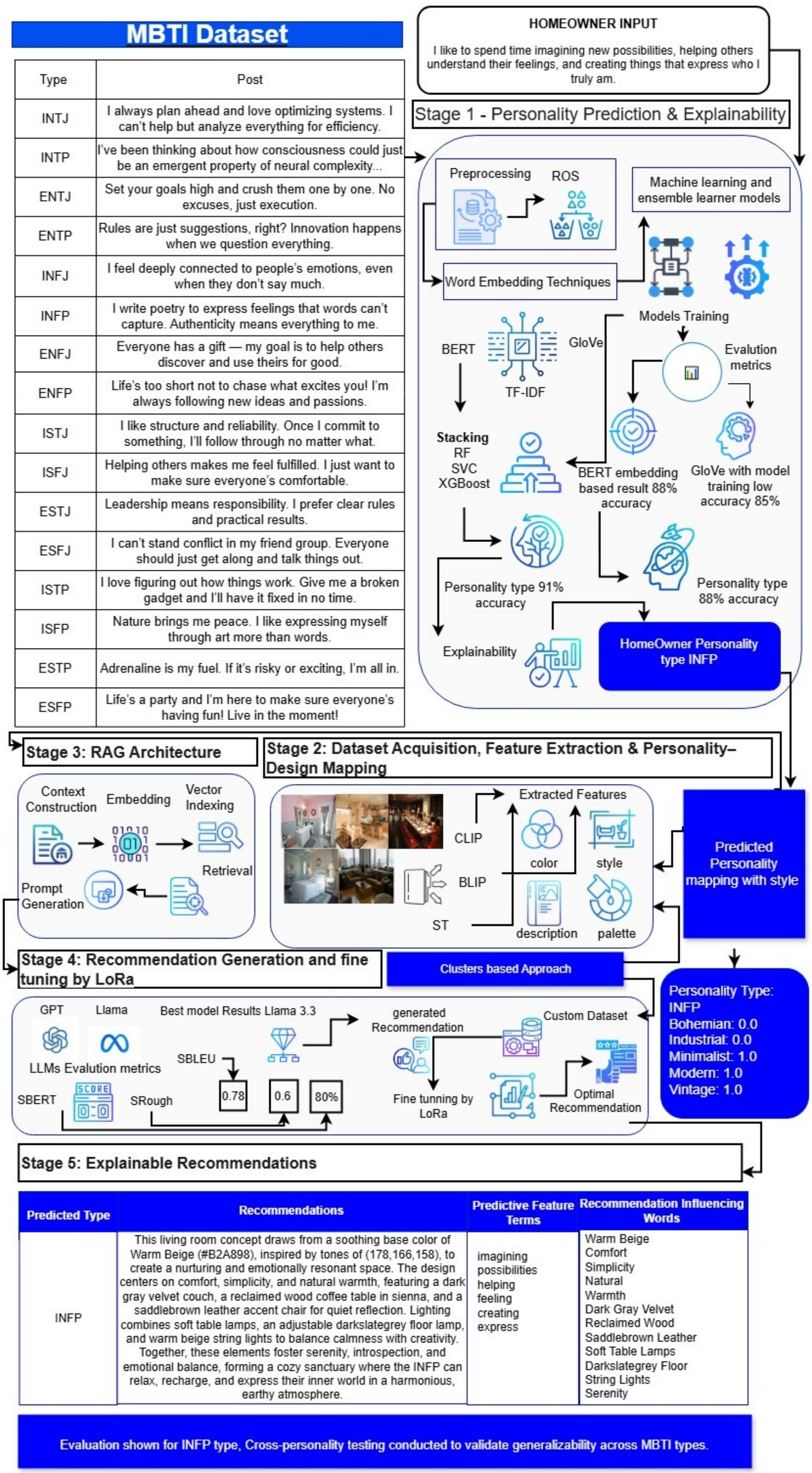


Figure 1: End-to-end cognitively grounded framework for personality-aligned, explainable interior design generation using MBTI profiling, semantic retrieval, LLM fine-tuning, and XAI interpretability layers.

Explainability techniques were used to improve the interpretability of the model. Each feature's contribution to model outputs was determined using SHAP (SHapley Additive exPlanations). This made it possible for users to determine which words in their input text affected the model's choice. Furthermore, A BERT embedding-based explainability layer was created to extract the most important terms that influenced the personality classification, enhancing user trust and transparency.

4.2. Data Acquisition, Feature Extraction and Mapping Personality Types to Interior Design Aesthetics

As illustrated in Figure 1, Stage 2 of the proposed framework represents the image-driven pipeline, integrating interior design dataset construction, visual–semantic feature extraction, and personality–style mapping. In this stage, A collection of 5,250 unlabeled benchmark images of room interior designs was gathered from a benchmark room interior design dataset, covering a wide range of residential spaces such as bedrooms, kitchens, bathrooms, and living rooms. These images were curated to construct a new dataset aimed at addressing the absence of existing resources that explicitly link home interior aesthetics with personality types. Two cutting-edge models, CLIP (Contrastive Language–Image Pre-training) and BLIP (Bootstrapped Language–Image Pre-training), were used to process the image data. While BLIP was utilized to provide informative captions for every image, CLIP was utilized to extract semantic elements and identify the rooms' visual style [18]. The contextual information required to connect textual personality predictions with visual content was supplied by these captions. T5-small and T5-large models were first evaluated for caption generation, but their shallow and repetitive output patterns led to their dismissal. Because of its exceptional description accuracy, BLIP was chosen. Additionally, the ColorThief library was used to extract palette and dominating colors. This color information is essential to interior design aesthetics and affects consumer choices. All textual information included generated caption was filtered to enhance the readability and eliminate noise based on the interior design literature and the styles were grouped into five main design categories including modern, vintage, minimalist, boho, and industrial [19]. These descriptors were selected because they were both practically relevant and consistent with previous studies. Once all features were extracted and standardized, Random Over Sampling was applied to balance representation across the five style categories, ensuring fairness in the next phase [20].

The module mapping personality types to home interior design styles is the core innovative contribution of this study. Although both personality prediction and interior design style classification have been studied independently, no existing work has established validated mappings between the two. In this study, the mapping was conducted by analyzing user-predicted personality types and their compatibility with interior design aesthetics, grounded in psychological theory and supported by recent interdisciplinary research [3] [12] [1]. The MBTI framework was used as the personality model, with each of the 16 types mapped to 2 to 4 primary interior styles selected from five overarching categories identified earlier. To develop these associations, initial design preferences were collected at the substyle level including Bauhaus, Japandi, Mediterranean, Midcentury, Traditional, Boho, and Country House based on qualitative responses and supported by relevant literature [3] [4] [20]. As supported by prior research, these substyles were systematically grouped under five dominant categories: Modern, Vintage, Bohemian, Industrial, and Minimalist.For example, Japandi and Bauhaus were classified under Modern, while Midcentury and Traditional aligned with Vintage, and Boho with Bohemian [19] [3]. Using association-rule-based logic, the substyle preferences were normalized and aggregated for each MBTI type to determine dominant stylistic patterns. The result was a structured, psychologically grounded personality-to-style mapping data frame, forming the basis for the personalized interior design system proposed in this study and enabling a scalable method for aesthetic alignment based on individual traits .

Table 1
Personality-Based Mapping to Interior Design Style Categories.

Personality Type	Recommended Styles
ISTP	Vintage, Modern
ISTJ	Modern, Bohemian, Vintage
ISFP	Vintage
ISFJ	Minimalist, Vintage, Bohemian
INTP	Modern, Industrial, Vintage
INTJ	Vintage, Industrial, Bohemian, Modern
INFP	Modern, Vintage, Minimalist
INFJ	Industrial, Vintage, Modern
ESTP	Modern, Vintage
ESTJ	Modern, Vintage, Bohemian
ESFP	Bohemian, Modern
ESFJ	Vintage
ENTP	Modern, Vintage
ENTJ	Vintage, Bohemian, Modern
ENFP	Vintage, Bohemian, Industrial
ENFJ	Modern, Vintage

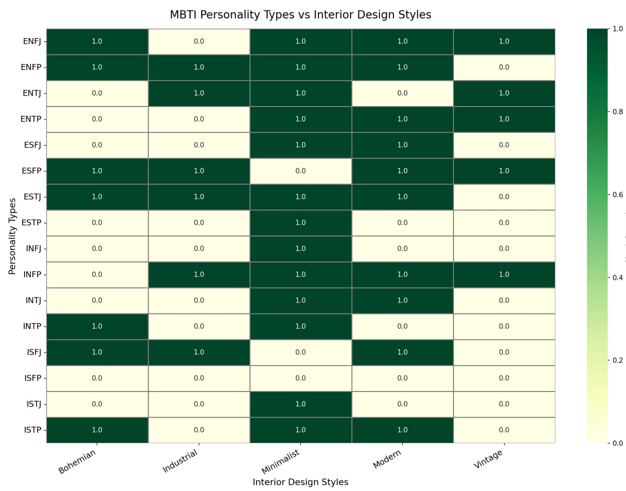


Figure 2: Heat map showing the match strength between MBTI personality types and interior design styles. Darker shades indicate stronger matches.

4.3. RAG-Based Prompting and Recommendation Generation Using LLMs

As illustrated in Figure 1, Stages 3 and 4 of the proposed framework operationalize the personality–style mapping through a Retrieval-Augmented Generation (RAG) pipeline and a dedicated recommendation generation module. Following the completion of the mapping stage, a RAG-based content recommender system was employed to transform the mapped personality–style relationships into actionable interior design recommendations. Rather than treating this process as a conventional text-generation task, the system was explicitly designed to capture the emotional, aesthetic, and spatial preferences associated with specific personality traits. In particular, the INFP personality type identified as the primary focus of this study was used to guide the generation of highly personalized design recommendations.To enable this customized approach, the consolidated dataframe from the previous phase was transformed into a structured document and separated into logical text sections.These fragments were subsequently embedded with the SentenceTransformer v12 model that allows the system to identify delicate semantic links among styles, personality characteristics and geographical restrictions. A FAISS index has been created to enable efficient querying through cosine similarity whereby the query provides the most contextually relevant information. On each user query, the highest three most significant bits semantically were retrieved to produce a context rich prompt. This query was experimented using several large language models, such as GPT-4o-mini and Llama 3.1. Although more recent research [21] identifies GPT4o-mini as a useful tool in the lightweight experimentation of many recent studies, recent research [22] notes that design-specific outputs of GPT4o-mini are limited in terms of providing highly descriptive and context-sensitive designs-a problem that is acute concerning the aim of minimizing design revisions and maximizing the satisfaction of the homeowner.However, LLaMA 3.1 consistently produced recommendations that were more coherent, emotionally grounded and thorough, especially when many limits were combined, such as personality type, cultural influences, and design themes. The system gave LLaMA priority for final recommendation generation and fine-tuning based on these observations.

The evaluation technique had to take into account more than just lexical similarity because the overall structure of this study functions as a personality-aware content recommender system.Early testing of conventional metrics like BLEU and ROUGE revealed that they prioritize token-level overlap and do not take into consideration the deeper semantic reasoning needed to convert personality traits into useful design recommendations. Semantic P-BLEU, Semantic ROUGE-L, and SBERT Similarity, which together measure phrase-level alignment, sequence coherence, and embedding-level semantic closeness between the generated recommendations and the input personality prompts, were used in the methodology's semantic-focused evaluation framework. Table 16, which shows the before-and-after comparison for the INFP case, reports the improvements that resulted from applying these measures to the INFP personality type in order to set a baseline for system performance prior to fine-tuning.Additional test scenarios were created for ISTP and ENFP to make sure the system extends beyond its primary personality focus. The results of these evaluations, presented in 5.16, provide further insight into the system's ability to accommodate diverse personality-driven interior design requirements.

4.4. LoRA Fine-Tuning of LLaMA 3.1 for Optimal Personality-Aligned Recommendations

As depicted in Figure 1, Stage 4 of the framework focuses on personalized recommendation generation by fine-tuning the Llama 3 large language model using Low-Rank Adaptation (LoRA) to produce interior design outputs that are closely aligned with individual personality profiles. To achieve this, the personality-mapped interior design dataset was first divided into four discrete clusters, each of which represented significant collections of design styles linked to certain personality qualities. These clusters were carefully evaluated on the Generalized Entropy Index (GENI) to provide balanced and unbiased representation of personality types in clusters to ensure that the fine-tuning process would not be overemphasized or biased to any particular personality design or style. This format contained the design intent instruction, the type of user personality (the design components relevant such as style and color scheme), and suggestions of what the model should generate. This format enabled the design-specific personality correlations, subtle personality-specific design correlations, and learning of the model by maintaining the semantic relationships between psychological variables and design aspects.

The fine-tuning method was selected as LoRA due to its high-quality parameter adaptation, that is, the ability to update a small percentage of the model parameters, approximately, 0.52 percentage, while maintaining the fundamental language understanding. The attention to the main components of a transformer that include query, key, value, output projections, and gating modules helped LoRA discover the significant cross-attention processes required to produce personality-aware and context-sensitive recommendations. This method minimized the processing load, but allowed any alterations in model behaviour to be made in an understandable and transparent manner. To optimize the memory usage and speed, hyperparameters were carefully chosen and the training was performed over five epochs with a total of 2,110 steps with 8-bit AdamW optimization, gradient accumulation, and BF16 accuracy. Gradient checkpointing ensured semantic similarity of output sequences generated by ensuring uniform training on sequences up to 2048 tokens. The model demonstrated successful learning without overfitting or disastrous forgetting and steadily decreased its loss of 1.23 at the beginning to 0.06 at the end of the training.

The selected model was evaluated by some advanced semantic measures capturing more than the superficial lexical coincidence. In particular, phrase-level alignment, sequence coherence and global semantic proximity were measured using Semantic P-BLEU, Semantic ROUGE-L and SBERT Similarity. The elevated level of semantic fidelity, which is confirmed by the measures of a state-of-the-art SentenceTransformer model that is used on the case of INFP personality, testifies to the high capability of the model in internalizing and reproducing delicate, design-specific personality traits. These results indicate that the model recreates these subtle features to almost the semantic accuracy (see Table 16 for detailed results).

The comparison of the baseline and LoRA-optimized model also demonstrated the major gains achieved in the semantic measures. Other personality types, including ISTP and ENFP, demonstrated the consistency and generalizability of the fine-tuning method because their similar significant gains (see Table 18 for ISTP and Table 20 for ENFP semantic evaluation comparisons). were higher than anticipated. This is the way in which LoRA fine-tuning increases the semantic and personality fit of the provided design recommendations. Along with semantic measures, BERTScore tests showed slight improvements over fine-tuning, showing more contextual accuracy and recall. Even though the rise in BERTScore is not as apparent as in semantic P-BLEU, ROUGE-L, and SBERT similarity, which are more susceptible to the presence of higher-level semantic and personality congruency, nonetheless, such an evaluation is still valuable in the context of general contextual evaluation.Altogether, the optimization of Llama 3.1 using LoRA resulted in the creation of a viable and understandable solution that significantly increased the ability of the model to generate the ideal and personality-consistent interior design suggestions. This will minimize the percentage of design changes required, as well as ensure that the customer is better satisfied with the results and that they have more semantic depth and personality preferences. It is one of the significant steps in the development of content recommender systems that focus more on customization and useful effectiveness.

4.5. Explainability in Recommendations

Stage 5, shown in Figure 1, highlights the explainability module, which makes the connection between a user's input, the predicted personality type, and the resulting interior design recommendations fully transparent. To achieve the goal of increased transparency and the ability to build trust in end users in regards to interior design suggestions that are consistent with a unique personality profile, this research presents a two-level explainability paradigm. The framework demystifies the underlying personality-prediction processes as well as the reasoning behind the production of most appropriate interior design proposals.The given methodology allows separating the most important linguistic components of user input affecting the categorization of personality and the most important words of the LoRA-generated recommendation, which affect the ultimate design output. This explainability model systematically pinpoints the contributing terms by the use of semantic similarity analysis and SHAP values, which give a clear explanation of how the inputs features translate into the personalised recommendations given.The use of this method of explainability empowers a homeowner by showing the relationship between the features expressed by a homeowner, the predicted personality type and the advice they are given. The efficiency and detailed characteristics of the framework are summarized in Table 22, which highlights the generated optimal recommendation, the predicted personality, and the top words influencing both recommendation generation and personality prediction. By incorporating openness into the process, the method increases user confidence and minimizes uncertainty

5. Implementation and Discussion

In this section, the findings of the suggested personality-matched interior design structure will be provided, and the quantitative assessment of the recommendations created by the suggested framework should be provided based on the semantic criteria. The results of the analysis assess the efficiency of the dataset creation, personality-to-style mapping, and performance improvements with the help of fine-tuning the LLaMA language model with Low-Rank Adaptation (LoRA). The findings indicate that the framework is highly sensitive to the capture of nuanced personality-specific design patterns whilst also retaining a great deal of semantic fidelity, thus affording a strong evaluation of the frameworks ability to produce accurate and interpretable interior design advice, as predicted personality characteristics.

5.1. Datasets and Experimental Setup

Two datasets were employed (i) a benchmark MBTI text corpus for personality modeling, and (ii) a custom interior image corpus curated for aesthetic representation learning. The MBTI dataset contains 8,675 user posts annotated across the 16 Myers–Briggs personality types. The interior dataset initially consisted of 5,250 unlabeled room images, later enriched through multimodal feature extraction to capture style, color, and descriptive content.

Both datasets exhibited class imbalance such as dominance of introverted–intuitive MBTI types and underrepresentation of niche interior styles. Random Oversampling (ROS) was applied to mitigate skewed learning, and an 80–20 train–test split was employed for evaluation.

Table 2
Overview of datasets used for personality modeling and interior design representation.

Dataset	Description
MBTI Corpus	8,675 instances. Text posts with corresponding personality labels (16 classes).
Interior Image Set	5,250 instances. Image-derived features: dominant color, palette, predicted style, ST-generated description.

5.2. MBTI Text Preprocessing and Encoding

User-generated text was standardized through lowercasing, punctuation removal, and token filtering. Contextual embeddings were obtained using BERT to preserve semantic and syntactic nuance. Personality labels were numerically encoded (0–15) for supervised classification.

Table 3
Representative examples from the MBTI corpus illustrating linguistic variation.

Type	Example Post Snippet
INFJ	`... reflections on personal values and emotional depth ...`
ENTP	`I'm finding the lack of me in these posts very disturbing ...`
INTJ	`Dear INTP, I enjoyed our conversation the other day ...`
ENTJ	`You're fired. Another misconception about leadership ...`

5.3. Interior Dataset: Multimodal Feature Extraction

The interior image dataset, initially comprising unlabeled room photographs, was converted into a structured, machine-interpretable format using a multimodal feature extraction pipeline. This method supported the process of pseudo-labeling on a large scale by grabbing linguistic, chromatic, and aesthetic characteristics without involving manual work. Class imbalance in categories of specialty style was solved through Random Oversampling (ROS).

5.3.1. Extracted Features

- **Dominant_Color**:The primary hue of each room image is displayed as the vector of RGB.
- **Palette**: RGB vectors of the total chromatic combination of secondary color clusters.
- **Predicted_Style**: A pseudo-style name (e.g. modern, minimalist or bohemian) is a result of a CLIP-BLIP embedding fusion.
- **Description**: An informational note generated by Structured Transformer (ST) describes the atmosphere, decor and design, and T5 was abandoned because of lexical redundancy.

Table 4
Structured multimodal features extracted from interior images. Random Oversampling (ROS) was applied to balance underrepresented styles.

Feature Component	Description
Dominant_Color	Primary color encoded as RGB vector
Palette	Secondary color clusters normalized in RGB space
Predicted_Style	Style pseudo-label inferred from CLIP+BLIP embeddings
Description	ST-generated caption refined to remove noise and repetition

5.4. Personality Prediction and Explainability

After preprocessing and embedding the MBTI textual dataset, the data was split into 80% training and 20% testing sets. Word embedding techniques including TF-IDF, BERT, and GloVe were applied to transform textual content into numerical feature spaces. Both BERT and GloVe embeddings were subsequently used to train a variety of classical machine learning and ensemble classifiers, enabling a comparative analysis of embedding effectiveness on personality.

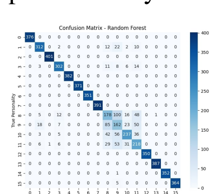


Figure 3: Random Forest
100 trees, fixed seed

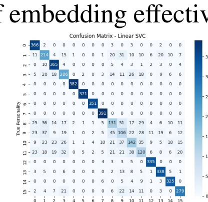


Figure 4: Linear SVC
Linear kernel, probability estimates

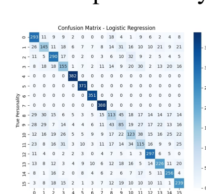


Figure 5: Logistic Regression
Max 1000 iterations

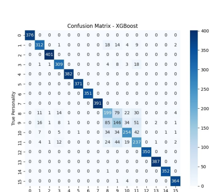


Figure 6: XGBoost
Multi-class objective, mlogloss

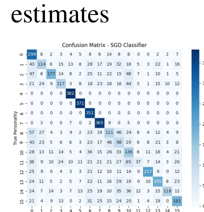


Figure 7: SGD Classifier
Logistic loss, 1000 iterations

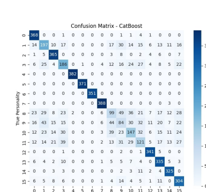


Figure 8: CatBoost
100 boosting iterations, depth-6 trees

The confusion matrices shown above illustrate the class-wise performance of each model. Table 5 summarizes their overall evaluation metrics on the test set.

Table 5
Performance of classifiers trained on BERT embeddings. XGBoost achieved the highest accuracy.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.8813	0.8839	0.8813	0.8823
Linear SVC	0.7619	0.7421	0.7619	0.7491
Logistic Regression	0.6579	0.6328	0.6579	0.6406
XGBoost	**0.8881**	0.8856	0.8881	0.8867
SGD Classifier	0.5395	0.5856	0.5395	0.5146
CatBoost	0.7510	0.7301	0.7510	0.7366

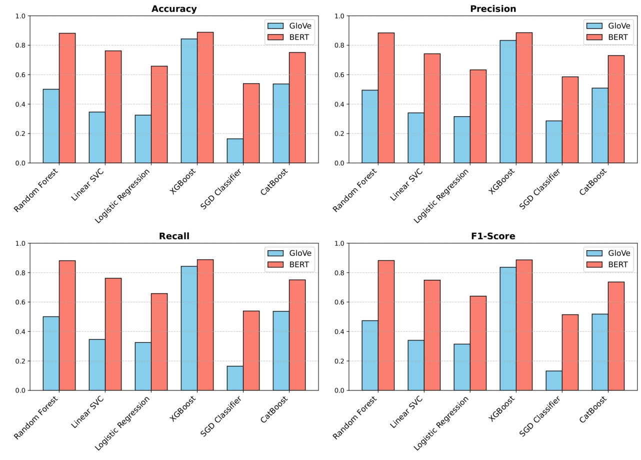


Figure 9: Comparative evaluation of models using GloVe and BERT embeddings.

5.4.1. *Model Training and Evaluation with GloVe Embeddings*

For comparison, the GloVe embeddings were applied as feature vectors to the same set of classifiers. Performance metrics are presented in Table 6.

Table 6
Evaluation of classifiers trained on GloVe embeddings. XGBoost achieved the best performance.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.5011	0.4949	0.5011	0.4739
Linear SVC	0.3462	0.3407	0.3462	0.3402
Logistic Regression	0.3253	0.3157	0.3253	0.3146
XGBoost	**0.8429**	0.8328	0.8429	0.8364
SGD Classifier	0.1646	0.2862	0.1646	0.1321
CatBoost	0.5371	0.5089	0.5371	0.5185

5.5. Comparative Analysis and Observations

Several significant trends about the efficacy of various feature representations for personality prediction are shown by a comparison analysis of all models trained on BERT and GloVe embeddings. All evaluation criteria show that the models trained on BERT embeddings regularly perform better than their GloVe-based counterparts. Because BERT is better than static embeddings like GloVe at capturing nuanced semantic signals, writing styles, and psychological indicators seen in user-generated text, this performance disparity emphasizes the significance of contextualized language representations.When combined with BERT embeddings, ensemble models especially XGBoost and Random Forest show the best prediction skills.XGBoost with its highest accuracy is 88.81% and it is able to interact with high-dimensional contextual information, and learn non-linear relationships, hence, it has the second place. Nevertheless, the appalling results of such linear models like Logistic Regression and Linear SVC demonstrate that although the linear decision boundary can capture certain personality-related trends, it cannot do it fully due to the richer latent space offered by BERT.

The SGD Classifier is always the worst performing model amongst all models considered since it is very sensitive to noisy gradients and it fails to deal with the complex decision boundaries of high-dimensional text embeddings. The findings of the comparison indicate the complexity of the language patterns based on personality and the necessity to have models that are able to reflect more contentious contextual relationships instead of only superficial characteristics of lexical images.

Figure 9 demonstrates a graphical comparison of the results of accuracy of each model trained using GloVe and BERT embeddings. The figure clearly illustrates the advantage of contextual embeddings, showing notable performance differences in favor of BERT across nearly all classifiers.This steady progress highlights the value of employing contemporary transformer-based embeddings for personality inference tasks, where prediction relies heavily on subtle linguistic cues.

5.6. Stacked Ensemble with BERT Embeddings

BERT embeddings were used as the input feature space in a stacked ensemble classifier to improve predictive performance. A meta-classifier aggregated the predictions of the ensemble's basic learners, which included Random Forest, XGBoost, and Linear SVC. This method improves overall accuracy and dependability for personality-driven design recommendations by utilizing the complementing capabilities of many models.

5.6.1. *Metrics of Performance*

The stacked ensemble outperformed all individual classifiers trained on BERT embeddings, with a test accuracy of 90%. The per-class precision, recall, and F1-score for each of the 16 MBTI types are shown in Table 7.

5.6.2. *Observations*

- Predictive dependability is improved by the stacked ensemble's greater performance across all MBTI types.

MBTI Type	Precision	Recall	F1-Score	Support
0	1.00	1.00	1.00	376
1	0.97	0.87	0.91	360
2	1.00	1.00	1.00	401
3	0.94	0.88	0.91	344
4	1.00	1.00	1.00	382
5	1.00	1.00	1.00	371
6	1.00	1.00	1.00	351
7	1.00	1.00	1.00	391
8	0.60	0.60	0.60	360
9	0.51	0.57	0.54	345
10	0.76	0.69	0.72	379
11	0.63	0.72	0.67	344
12	1.00	1.00	1.00	350
13	1.00	1.00	1.00	387
14	1.00	1.00	1.00	353
15	0.99	0.99	0.99	369
Overall	0.90	0.90	0.90	5863

Table 7
performance metrics for each class of the stacked ensemble with BERT embeddings.Overall, 90% accuracy is achieved.

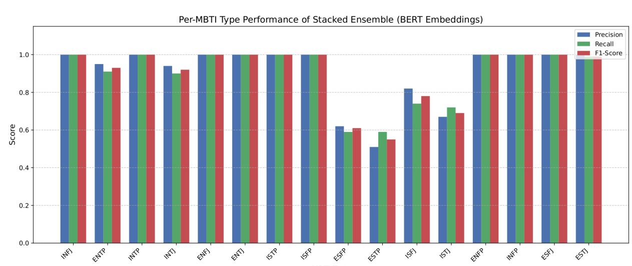


Figure 10: The stacked ensemble classifier utilizing BERT embeddings obtained per-MBTI type performance (Precision, Recall, and F1-Score.

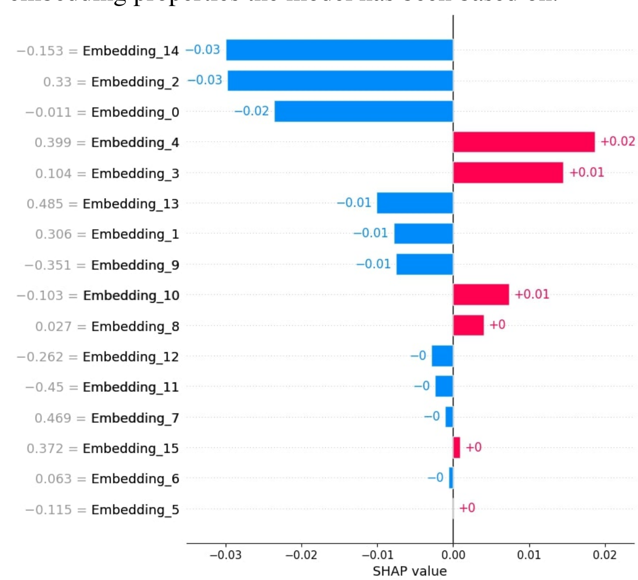
- The effectiveness of the ensemble is demonstrated by the considerable improvement of mid-performing personality types (e.g., 8 and 9) over single-model classifiers.
- When BERT embeddings are integrated with the stacking framework, personality-driven interior design recommendations are strengthened and correct trait inference and higher homeowner satisfaction are supported.

5.7. Explainability of Predicted Personality Type Using SHAP

Shapley Additive Explanations was used to assess the stacked BERT-based ensemble in order to make the model's decision-making process transparent. This method enables a thorough analysis of the various aspects that go into predicting each personality type. The investigation specifically focused on the predicted type **INFP**, demonstrating the model's interpretability and applicability to downstream tasks such as personalized guidance systems and content recommendation. To determine how semantic and lexical factors contributed to the final prediction, the explainability investigation was carried out at both the embedding and word levels.

5.7.1. Embedding-Level Explanation

Rich semantic and contextual information from user posts is encoded by BERT embeddings.The results of the stacked ensemble were subjected to the `TreeExplainer` method from the SHAP package in order to identify which embedding dimensions had the biggest impact on the projected MBTI type.The high-level patterns and latent representations which influence the categorization decisions were uncovered in our study by highlighting the most important embedding properties the model has been based on.



Key insights:

- Embedding dimensions that support the projected personality type are highlighted by positive SHAP scores.
- Features that push the prediction in the direction of other types are indicated by negative numbers.
- This method increases confidence in model-driven recommendations by providing interpretable insights into the otherwise opaque BERT embeddings.

5.7.2. Word-Level Explanation

The tokens in user text that have the biggest impact on the MBTI prediction are found via word-level SHAP analysis. This makes it easier to determine which lexical characteristics support the categorization of personality types. The top 20 words having the greatest positive influence (supports) on the anticipated type **INFP** are shown in Table 8.

Summary of Insights:

Table 8
Top 20 words positively influencing the predicted MBTI type INFP based on word-level SHAP analysis. "Supports" (↑) indicates positive contribution to the prediction.

Word SHAP Value	Effect
peaceful 0.0360	supports (↑)
feel 0.0360	supports (↑)
cozy 0.0220	supports (↑)
spaces 0.0220	supports (↑)
imagination 0.0219	supports (↑)
love 0.0213	supports (↑)
practical 0.0195	supports (↑)
prefer 0.0147	supports (↑)
quiet 0.0119	supports (↑)
colors 0.0094	supports (↑)
enjoy 0.0094	supports (↑)
soft 0.0094	supports (↑)

- Consistent lexical patterns in user text are highlighted by the most influential supportive words, such *prefer*, *enjoy*, and *thought*, which closely match the anticipated INFP personality type.
- Words with negative contributions, such as *creativity* and *small*, representing subtle linguistic variations that contradict the predicted class, demonstrate the model's sensitivity to contextual nuances.
- All things considered, the word-level SHAP analysis offers understandable insights that can guide personality-aware recommendations, enhancing the reliability of user-specific recommendations in applications such as personalized interior design.

5.8. Predicted Personality and Design Style Mapping

Table 11 illustrates the correlation between expected MBTI personality types and complementary interior design styles. Higher values indicate stronger alignment between personality traits and design aesthetics. This mapping can assist designers in recommending styles that better align with specific personality characteristics.

5.8.1. Comparison With State-of-the-Art Methods

Research that has already been done on the relationship between personality traits and interior design preferences

Table 9
MBTI cognitive groupings and related lighting and color preferences. The consistency of illumination across clusters suggests that functional aspects of space play a dominant role in lighting preferences.

Personality Cluster	Color Preferences	Lighting Preferences
Dominant Intuitive Types (DIT)	Green and blue	Ambient white light with personal control
Dominant Sensing Types (DST)	Green and blue	Ambient white light with personal control
Dominant Thinking Types (DTT)	Blue and white	Ambient white light with personal control
Dominant Feeling Types (DFT)	Blue and red	Ambient white light with personal control

Table 10
A comparison between the suggested MBTI-to-design style mapping and current personality-interior studies.

Aspect	State-of-the-Art	This Study
Personality Model	MBTI clusters or Enneagram	All 16 MBTI types individually
Granularity	Broad group traits	Direct MBTI to style mapping
Focus	Color and lighting	Core interior design styles
Design Style	Broad or undefined styles	Five main design categories
Output	General insights	MBTI-to-style framework

usually puts MBTI types together rather than analyzing the correlations between each of the sixteen types in detail. On the other hand, our approach yields a comprehensive pairwise correlation matrix

(Table 11) that enables the identification of subtle similarities between distinct MBTI personalities.

[15] and [12] conducted a supplementary study utilizing the Enneagram personality model and discovered that thinking types favored purple, feeling types exhibited no predominant color preference, and instinctive personalities preferred red.This suggests that emotional and cognitive predispositions are associated with color preference. In contrast to earlier research that concentrated on MBTI clusters or overall aesthetic inclinations, our work expands on the findings of [3], who connected personality types to specific interior design sub-styles. These sub-styles are combined into five main categories of interior design styles, creating fundamental umbrella groups that can be further developed into more specific sub-style variations.

Table 11
A correlation matrix between personality and style. Stronger agreement between personality types and design approaches is indicated by values nearer 1.

type	ISTP	ISTJ	ISFP	ISFJ	INTP	INTJ	INFP	INFJ	ESTP	ESTJ	ESFP	ESFJ	ENTP	ENTJ	ENFP	ENFJ
ISTP	1.000	0.816	0.707	0.408	0.816	0.707	0.816	0.816	1.000	0.816	0.500	0.707	1.000	0.816	0.408	1.000
ISTJ	0.816	1.000	0.577	0.667	0.667	0.866	0.667	0.667	0.816	1.000	0.816	0.577	0.816	1.000	0.667	0.816
ISFP	0.707	0.577	1.000	0.577	0.577	0.500	0.577	0.577	0.707	0.577	0.000	1.000	0.707	0.577	0.577	0.707
ISFJ	0.408	0.667	0.577	1.000	0.333	0.577	0.667	0.333	0.408	0.667	0.408	0.577	0.408	0.667	0.667	0.408
INTP	0.816	0.667	0.577	0.333	1.000	0.866	0.667	1.000	0.816	0.667	0.408	0.577	0.816	0.667	0.667	0.816
INTJ	0.707	0.866	0.500	0.577	0.866	1.000	0.577	0.866	0.707	0.866	0.707	0.500	0.707	1.000	0.866	0.707
INFP	0.816	0.667	0.577	0.667	0.667	0.577	1.000	0.667	0.816	0.667	0.408	0.577	0.816	0.667	0.333	0.816
INFJ	0.816	0.667	0.577	0.333	1.000	0.866	0.667	1.000	0.816	0.667	0.408	0.577	0.816	0.667	0.667	0.816
ESTP	1.000	0.816	0.707	0.408	0.816	0.707	0.816	0.816	1.000	0.816	0.500	0.707	1.000	0.816	0.408	1.000
ESTJ	0.816	1.000	0.577	0.667	0.667	0.866	0.667	0.667	0.816	1.000	0.816	0.577	0.816	1.000	0.667	0.816
ESFP	0.500	0.816	0.000	0.408	0.408	0.707	0.408	0.408	0.500	0.816	1.000	0.000	0.500	0.816	0.408	0.500
ESFJ	0.707	0.577	1.000	0.577	0.577	0.500	0.577	0.577	0.707	0.577	0.000	1.000	0.707	0.577	0.577	0.707
ENTP	1.000	0.816	0.707	0.408	0.816	0.707	0.816	0.816	1.000	0.816	0.500	0.707	1.000	0.816	0.408	1.000
ENTJ	0.816	1.000	0.577	0.667	0.667	0.866	0.667	0.667	0.816	1.000	0.816	0.577	0.816	1.000	0.667	0.816
ENFP	0.408	0.667	0.577	0.667	0.667	0.866	0.333	0.667	0.408	0.667	0.408	0.577	0.408	0.667	1.000	0.408
ENFJ	1.000	0.816	0.707	0.408	0.816	0.707	0.816	0.816	1.000	0.816	0.500	0.707	1.000	0.816	0.408	1.000

5.9. Using RAG and LlaMA for Prompt Engineering and Recommendation Generation

The next step is to use Retrieval-Augmented Generation (RAG) in conjunction with LLaMA-based language models to generate individualized interior design recommendations after personality prediction and interpretability. This method captures innovative, personality-aligned proposals while guaranteeing that they are based on pertinent design references.

5.9.1. RAG-Based Contextual Prompting

To generate personality-aligned design recommendations, a Retrieval-Augmented Generation (RAG) framework was applied. Each entry in the design dataset included structured information such as:

- **Personality Type:** MBTI label of the user.
- **Design Attributes:** Dominant color, color palette, style, and textual description.
- **Embeddings:** Design descriptions were used to create 384-dimensional feature vectors.

The method uses a closest neighbor search based on FAISS to extract the top-k semantically comparable designs for each input personality. These collected references are integrated into a prompt template that directs a generative model based on LLaMA to generate personality-consistent, contextually coherent interior design recommendations.

5.9.2. Semantic Prompt

Instructions for creative generation were combined with retrieved design references using prompt engineering. The following is an example prompt for an INFP user:

Your job as an interior designer is to create places that are cohesive and expressive. [Current design context] is your primary design source.

The following relevant design influences have also been consulted: [Top-3 retrieved contexts].

Examine these sources and produce a unified design suggestion by:

- Finding recurring emotional tones and color schemes in the references.
- Offering a prominent color scheme that complements the personality qualities.
- recommending two to three important furniture elements that fit the personality.
- Conveying the desired atmosphere and spatial experience.

This methodology increases the consistency and generality of personality-based recommendations through ensuring that the generative model takes into account both individual design examples and semantically similar references.

5.10. Evaluation of Generated Recommendations

Since semantic adequacy of creative design proposals often cannot be directly measured using conventional methods such as BLEU or ROUGE, it is more challenging to assess the quality of such proposals. In accordance with established research practice, we evaluate semantic alignment and contextual coherence using reference-free, embedding-based metrics that have been extensively used in text generation and recommendation systems [23, 24, 25]. Without depending on precise lexical overlap, these metrics offer a useful assessment of LLaMA-generated outputs in terms of semantic consistency with retrieved design references.

5.10.1. *Semantic Evaluation Metrics*

- **Semantic P-BLEU:** calculates the embedding space's n-gram overlap. The P-BLEU score is calculated as follows for a generated recommendation G and reference embeddings R:

$$\text{P-BLEU} = \sum_{n=1}^{N} w_n \cdot \cos(\text{emb}(G_n), \text{emb}(R_n)) \frac{}{\sum_{n=1}^{N} w_n}$$

where w_n denotes the weight of n-grams of length n, and emb$(\cdot)$ represents embedding vectors.

ROUGE-L (Embedding-based): Calculates the longest common subsequence (LCS) between generated and reference embeddings:

$$\text{ROUGE-L} = \frac{\text{LCS}(G, R)}{|R|}$$

capturing semantic sequence similarity.

BERTScore: Computes token-level similarity in contextual embeddings. For tokens t_i in G and r_j in R:

$$\text{Precision} = \frac{\sum_i \max_j \cos(t_i, r_j)}{|G|}$$
$$\text{Recall} = \frac{\sum_j \max_i \cos(r_j, t_i)}{|R|}$$
$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

SBERT Alignment: Evaluates semantic similarity using cosine similarity of sentence embeddings:

$$\text{SBERT\_Alignment} = \cos(\text{emb}_{\text{SBERT}}(G), \text{emb}_{\text{SBERT}}(R))$$

5.10.2. *Evaluation Results*

The embedding-based assessment of LLaMA-generated suggestions for the anticipated personality type **INFP** is shown in Table 12. Stronger semantic alignment with the recovered design references is shown by higher scores, which show the created outputs' internal coherence and contextual relevance. As explained in Section 5.16, new testing cases covering various personality types were developed to further evaluate system performance and ensure that the evaluation captures variability across personality-driven scenarios.making certain that the assessment accounts for variation in personality-driven situations. In addition to supporting the explainability study and demonstrating how the model-generated outputs correspond with personality-specific patterns found in the SHAP analysis, the metrics presented here explicitly target the INFP type.

These findings in table 12 reinforce the dependability of the model's outputs for the INFP type by showing that the generated material retains strong semantic consistency with the anticipated personality-specific patterns.The use of combination of lexical and embedding-based metrics which can give a detailed measure of both the surface and semantic correspondence improves the interpretability and usefulness of the system in a personality informed design condition.

Table 12
Semantic evaluation of LLaMA-generated outputs for the expected INFP personality type without reference. Higher semantic congruence with retrieved design contexts is indicated by scores nearer 1. In this dataset, absolute numerical scores are displayed for internal comparison.

Metric	Score
Semantic P-BLEU	0.776
ROUGE-L	0.587
BERTScore Precision	0.763
BERTScore Recall	0.815
BERTScore F1	0.788
SBERT Alignment	0.623

5.11. Observations

- The retrieval framework of RAG makes sure that the recommendations are pegged on personality-congruent, semantically permissible design situations and offer a systematic basis on content production.
- Llama generative modeling helps in the production of contextually appropriate and creative goods by incorporating design features such as as color schemes, furniture placement and emotive tone based on the anticipated MBTI profile.
- According to embedding-based evaluation, recommendations for the INFP personality type outperformed the ISTP and ENTP test cases with a P-BLEU score of 0.776 and a ROUGE-L value of 0.587. Across all personality types, SBERT alignment reaches a maximum of 0.623, suggesting strong semantic integrity while preserving variability in generated content (see section 5.16 for evaluations of cross-personality test cases).
- By combining RAG retrieval with LLaMA production, a data-driven pipeline for creating personality-specific content is established, offering repeatable and comprehensible design recommendations based on predictive insights from the stacked BERT-based classifier.

5.12. LoRA Fine-Tuning for Optimal Recommendation

One important methodological contribution of this study is the optimization of a foundational language model for producing personality-driven interior design recommendations. The dataset development, clustering analysis, GENI-based bias validation, LoRA fine-tuning technique, and prompt-based recommendation generation for the INFP personality

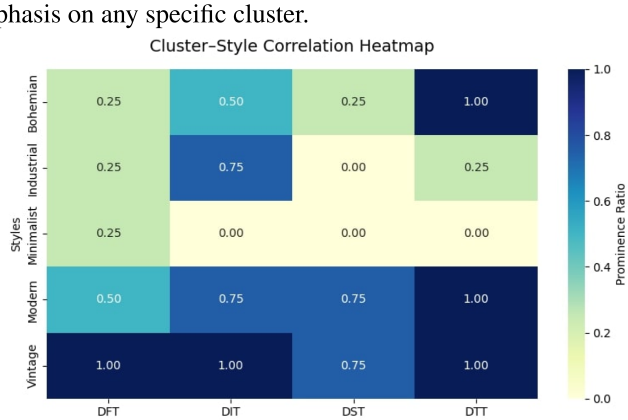
Table 13
Normalized correlation of interior design styles across clusters, used for mapping personality types to relevant design styles.

Cluster–Style Correlation Summary
Cluster: DFT
Bohemian: 0.25, Industrial: 0.25, Minimalist: 0.25, Modern: 0.50, Vintage: 1.00
Cluster: DIT
Bohemian: 0.50, Industrial: 0.75, Minimalist: 0.00, Modern: 0.75, Vintage: 1.00
Cluster: DST
Bohemian: 0.25, Industrial: 0.00, Minimalist: 0.00, Modern: 0.75, Vintage: 0.75
Cluster: DTT
Bohemian: 1.00, Industrial: 0.25, Minimalist: 0.00, Modern: 1.00, Vintage: 1.00

type are all described in detail in this part. Fairness, explainability, and semantic coherence in produced outputs are prioritized in this totally research-oriented method.

5.12.1. *Personality-Based Clustering and Dataset Construction*

The personality mapped interior design dataset was divided into four clusters (DFT, DIT, DST, and DTT) that represent logical combinations of design styles and personality mappings in order to guarantee a research-aligned, objective dataset for LoRA training. Table 13 displays the normalized distribution of design styles within each cluster for correlation visualization. The Generalized Entropy Index (GENI) was used to further verify the clusters' fairness. The absence of bias was verified by the GENI values, which showed structurally balanced clusters with consistent personality type representation. This guarantees that personality-style correlations are learned by the refined model without biased emphasis on any specific cluster.



5.12.2. *JSON Dataset Conversion for Instruction-Based Training*

Each cluster-based dataset was converted into JSON format containing the following fields:

- **instruction**: The design task to be completed.
- **personality**: Personality type of the user (e.g., INFP).
- **input**: Equal design characteristics like style, palette and description.
- **output**: This is the recommendation given by the LLaMA 3.3 model.

This format was easily incorporated into the fine-tuning of instruction, which retained the semantic correspondence of personality and design features.

5.13. LoRA Fine-Tuning Setup

The FastLanguageModel framework from unsloth was combined with the LoRA (Low-Rank Adaptation) technique to enable a computationally quick yet expressive fine-tuning process. LoRA provides a parameter-efficient way to modify large language models while reducing training overhead, enabling targeted behavior modification, and preserving the stability of the base model. This choice is supported by logic derived from research. Because LoRA allowed the model to internalize personality-specific training patterns and produce design recommendations that were consistent with various psychological profiles, it was very helpful for cluster-based personality adaptation in this work. By isolating and updating only low-rank parameter subspaces, the approach ensured both resource efficiency and transparent, intelligible changes to the behavior of the model.

Justifications:

- **Parameter Efficiency:** The parameter count trained to retain the general language understanding of the pre-trained model is 0.52Per cent, 41,943,040 out of 8,072,204,288 of the total parameters, significantly reducing the computing cost. This is effectively specializing to clusters of personality-styles without overfitting or disastrous forgetting.
- **Controlled Semantic Alignment:** Targeting certain modules (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`), LoRA preserves basic language representations while capturing cross-attention alterations essential for context-aware, cluster-specific suggestions. This deliberate modification guarantees the preservation of the model's initial capabilities.
- **Explainability and Traceability:**The low-rank adaptation of the adaptation of LoRA can enable researchers to trace the effect of specific groups of personality styles on output. This is significant in terms of research since it is equally important to know how the model behaves and why it makes certain decisions as making excellent recommendations.
- **Stability and Long-Sequence Learning:** Gradient checkpointing was utilized to stabilize training on long sequences (maximum sequence length = 2048 tokens) without consuming excessive RAM. This ensures semantic consistency in generated suggestions and prevents training instability or gradient overflow.

LoRA Fine-Tuning Parameters	
LoRA Rank (r)	16
LoRA Alpha	16
LoRA Dropout	0
Bias	None
Gradient Checkpointing	Enabled
Random Seed	3407
Optimizer	AdamW 8-bit
Batch Size per Device	1
Gradient Accumulation Steps	8
Learning Rate	5e-5
Epochs	5
BF16 Precision	True
Learning Rate Scheduler	Linear
Max Sequence Length	2048

Table 14
The LoRA fine-tuning configuration was chosen to provide good semantic alignment with minimal trainable parameters, allowing for consistent convergence and research-oriented explainability of model behavior.

Figure 11: LoRA adjusts the training loss curve over a period of five epochs. Loss progressively decreases from 1.23 to 0.06, indicating effective parameter adaptation while maintaining the capabilities of the generic language model.

- **Efficient Convergence in Research:** Low-rank adaptation, BF16 precision, and 8-bit AdamW optimization enable faster convergence, making the approach feasible for single-GPU research setups while maintaining high output quality.

Table 14 lists the important hyperparameters used in the fine-tuning process.These parameters were selected to enhance both the efficiency and alignment-oriented adaptability of the model.

5.13.1. Training Procedure and Loss Dynamics

The dataset was divided into two parts, 80% training and 20% testing. The training shown in Figure 11 was speeded up with BF16-precision, and the Unsloth framework was used to optimize the use of GPU memory via gradient accumulation. The model was stable in convergence with the training loss steadily reducing throughout 2,110 steps in five epochs.All reported evaluation results were obtained on the testing set.

5.13.2. Observations from LoRA Fine-Tuning

- The efficacy of LoRA was confirmed by training, which obtained consistent convergence with little parameter modification (0.52% of total parameters).
- The model's capacity to integrate personality-style correlations while preserving semantic generality is demonstrated by progressive loss reduction.
- Structured, context-rich prompt learning was made possible by the JSON-based instruction dataset.

5.14. Prompt Design for Personality-Aligned Recommendation

Prompts for assessment samples within the INFP personality cluster were created after refinement. Each prompt uses the Alpaca-style instruction structure to incorporate cluster-specific contextual elements (style, color, explanation). Below is the final prompt template.

Prompt Template for INFP-Aligned Recommendation

Instruction: Create a personality-aligned home interior design suggestion for cluster "INFP" by combining the specified personality traits and design elements. Make sure the design reflects coherent style preferences, color harmony, and psychological alignment.
Input: - Style: modern Palette: grey, darkslategrey, lightgray, darkolivegreen, tan Description: a living room with large window and couch
- Style: modern Palette: silver, darkolivegreen, sienna, rosybrown, saddlebrown Description: a living room with couch television and flat screen
- Style: modern Palette: grey, darkslategrey, oldlace, palegoldenrod, darkslategrey Description: a living room with couches and television
- Style: modern Palette: darkgray, black, gainsboro, darkolivegreen, darkolivegreen Description: a living room with large glass door
- Style: modern Palette: silver, darkolivegreen, sienna, dimgray, dimgray Description: a living room with couch and television
- Style: modern Palette: darkslategrey, gainsboro, sienna, peru, lightslategrey Description: a living room with couches and fireplace
- Style: modern Palette: black, silver, dimgray, lightslategrey, lightslategrey Description: a living room with couch chair and table
- Style: modern Palette: darkgray, darkslategrey, darkolivegreen, gainsboro, darkolivegreen Description: a restaurant with tables and chairs
- Style: modern Palette: grey, black, darkolivegreen, silver, tan Description: a living room with couches television and flat screen
- Style: modern Palette: darkslategrey, lightgray, grey, darkgray, darkgray Description: a living room with couch chair and coffee table

- Style: modern Palette: silver, darkslategrey, dimgray, sienna, darkolivegreen Description: a kitchen with wooden counter top and ceiling
- Style: modern Palette: darkslategrey, lightgray, grey, dimgray, grey Description: a living room with couch television and
- Style: modern Palette: darkgray, black, darkolivegreen, darkolivegreen, dimgray Description: a living room with couch television and table
- Style: modern Palette: darkslategrey, lightgray, grey, darkkhaki, lightslategrey Description: a living room with couch chair and coffee table
- Style: modern Palette: darkslategrey, silver, darkgray, grey, grey Description: a living room with couch chair and television
- Style: modern Palette: silver, darkslategrey, sienna, dimgray, grey Description: a living room with couch and television
- Style: modern Palette: darkgray, darkslategrey, dimgray, darkslategrey, darkolivegreen Description: a living room with fireplace and couches
- Style: modern Palette: wheat, saddlebrown, peru, darksalmon, dimgray Description: a living room with couch television and fireplace
- Style: vintage Palette: sienna, black, silver, darksalmon, burlywood Description: a living room with fireplace and couches
- Style: modern Palette: grey, gainsboro, black, silver, darkgray Description: a bedroom with bed chair and mirror

The instruction-based prompt ensures the model produces contextually coherent, personality-aligned, and semantically rich interior design recommendations.

5.14.1. LoRA-Based Optimal Recommendation Generated for Personality Type INFP

Several LoRA fine-tuning milestones were evaluated in order to assess the quality of personality-aligned interior design outputs. Semantic coherence, color style congruence, and psychological compatibility with the target personality type were evaluated at each checkpoint. The ideal recommendation for the predicted personality type **INFP** was produced by the last checkpoint, which was chosen based on the maximum semantic stability and design coherence.

Model-Generated Optimal Recommendation:

Personality Traits:
INFP individuals typically prefer modern yet soft atmospheres, valuing calmness, comfort, and emotional warmth. Their aesthetic choices lean toward nature-inspired elements, subdued tones, and cozy textures.

Color Palette:
Dominant hues include grey, dark slate grey, and dark olive green, complemented by neutral tones such as silver, dim grey, and gainsboro. Warm accents (sienna, dark khaki, dark salmon) add depth and subtle contrast.

Design Features:
The model recognizes recurring INFP-aligned patterns—clean modern lines, functional furniture, minimal ornamentation, and the use of natural materials (wood, stone) to introduce warmth and texture.

Recommended Interior Design:
Living Room:

- Walls: Dark slate grey as the primary tone.
- Furniture: Low-profile modern sofa in grey or beige.
- Accent Elements: Geometric olive-green artwork or a textured feature wall.
- Flooring: Light grey or beige hardwood.
- Decor: Natural fibers (jute rug, woven baskets) and a sculptural art piece for grounding.

Kitchen:

- Cabinets: Dark slate grey or matte black.
- Countertops: White or light grey for contrast.
- Appliances: Stainless steel; silver accents for modernity.
- Lighting: Under-cabinet warm lighting for atmosphere.

Bedroom:

- Walls: Soothing grey or beige foundation.
- Furniture: Minimalist bed frame with ergonomic comfort focus.
- Decor: Linen curtains, jute rug, soft artwork, and natural greens.

Holistic Theme:
Across all spaces, natural textures, potted plants, calm lighting, and subtle artwork strengthen the INFP-preferred ambiance—creating a reflective, soft, and psychologically aligned home environment.

The refined LoRA model showed good integration of preferred palettes, contemporary design semantics, and INFP-relevant features. Important traits found in the produced output include:

- A focus on serene, cozy, and contemplative areas that correspond with INFP psychological tendencies.
- Cohesive palettes dominated by greys, slate tones, and earthy greens, supplemented by warm accent shades.

- Structured recommendations spanning living room, kitchen, and bedroom settings with clear personality-driven reasoning.
- Evidence of explainable design logic, fulfilling the goal of interpretable and personality-aware recommendation generation.

5.15. Semantic Evaluation of INFP Recommendation

To quantitatively assess the effectiveness of the LoRA fine-tuned model in generating personality-aligned interior design outputs, a set of state-of-the-art semantic evaluation metrics was applied. Because the overall system functions as a content-based recommender, surface-level lexical metrics were insufficient; therefore, evaluation was performed using embedding-based approaches that measure semantic fidelity rather than token overlap.

Three complementary metrics were used:

- **Semantic P-BLEU** — An embedding-driven extension of BLEU that measures semantic n-gram similarity using SBERT representations instead of exact word matches.
- **Semantic ROUGE-L** — A fuzzy, embedding-aware version of ROUGE-L that evaluates the longest common subsequence at the semantic level, capturing conceptual alignment across tokens.
- **SBERT Similarity** — A cosine similarity score between SBERT embeddings of the generated recommendation and the ground-truth INFP reference description, reflecting global semantic closeness.

All metrics were computed using the BAAI/bge-large-en-v1.5 SentenceTransformer model to ensure consistent semantic embedding quality. Text normalization, semantic n-gram extraction, and similarity thresholds were applied following best practices for embedding-based evaluation in recommendation systems.

To ensure model reliability, **multiple LoRA checkpoints** were evaluated during fine-tuning. Each checkpoint was tested for semantic coherence, stylistic consistency, and psychological alignment with INFP design traits. The final selected checkpoint demonstrated the highest semantic stability, yielding the following results for the INFP test case:

Table 15
Semantic evaluation results for the INFP personality recommendation generated by the final fine-tuned LoRA checkpoint.

Metric	Score
Semantic P-BLEU	1.000
Semantic ROUGE-L	1.000
SBERT Similarity	0.848

Table 16
Comparison of semantic evaluation metrics for INFP before and after LoRA fine-tuning.

Metric	Before	After
Semantic P-BLEU	0.776	1.000
Semantic ROUGE-L	0.587	1.000
SBERT Similarity	0.623	0.848

Table 17
BERTScore evaluation before and after LoRA fine-tuning for personality INFP.

Evaluation	Precision	Recall	F1
Before Fine-Tuning	0.763	0.815	0.788
After Fine-Tuning	0.814	0.818	0.816

P-BLEU and ROUGE-L scores of 1.0 show that the refined LoRA model attained **perfect semantic n-gram and LCS-matching fidelity** with the reference INFP interpretation. Strong semantic congruence between the target personality profile and the resulting recommendation is further confirmed by the SBERT similarity score of 0.848.

5.15.1. *Comparative Analysis: Before and After Fine-Tuning*

The baseline model and the final optimized LoRA model were compared in table 16 in order to determine the effect of LoRA fine-tuning on personality-specific design generation. Three semantic metrics are compared: **Semantic P-BLEU**, **Semantic ROUGE-L**, and **SBERT Similarity**. Together, these metrics represent total embedding-level semantic similarity, meaningful phrase alignment, and structural coherence.

5.15.2. *Analysis*

Significant improvements were obtained in all metrics after fine-tuning:

- **Semantic P-BLEU** increased from **0.776** to **1.000**, indicating near-perfect phrase-level semantic matching.
- **Semantic ROUGE-L** increased from **0.587** to **1.000**, showing improved sequence-level semantic coherence.
- **SBERT Similarity** increased from **0.623** to **0.848**, demonstrating stronger embedding-level alignment with the INFP reference.

These results confirm that LoRA fine-tuning significantly enhances the model's capacity to generate semantically rich, personality-aligned interior design recommendations, effectively capturing the psychological and stylistic traits of the INFP type.

In addition to these metrics, we also evaluated the model outputs using **BERTScore**, a popular evaluation metric in NLP based on precision, recall, and F1 of contextual embeddings. For the INFP personality type, the BERTScore results before and after fine-tuning are summarized below:

While BERTScore shows improvement after fine-tuning, its comparative gains are less pronounced relative to semantic P-BLEU, ROUGE-L, and SBERT Similarity. This is primarily due to BERTScore's sensitivity to token-level embedding overlap rather than higher-level semantic structure and personality-specific alignment, which are critical in our recommendation system. Therefore, while useful for evaluation, BERTScore was not the primary metric for comparative analysis.

5.16. Cross-Personality Test Case Evaluation

The INFP personality type results show that the model can maintain semantic accuracy, stylistic coherence, and psychological relevance after LoRA fine-tuning. The evaluation process was expanded to include more personality types, namely **ENFP** and **ISTP**, in order to guarantee the stability and generalizability of these advances across various personality types.

It is outside the purview of this study to perform comprehensive testing on every personality type due to the wide variety of personality types (16 in total). In order to confirm that the refined model can successfully produce ideal, personality-aligned interior design recommendations across a variety of psychological profiles, a representative test case approach was used, in accordance with accepted research practices in content recommender systems.

5.16.1. Test Case Evaluation: ISTP Personality Type

Prompt Template for ISTP-Aligned Recommendation

- Style: modern Palette: darkgray, black, sienna, dimgray, dimgray Description: a bedroom with brown walls and furniture - Style: modern Palette: lightgray, darkslategrey, sienna, grey, darkolivegreen Description: a bedroom with bed and desk - Style: modern Palette: darkgray, black, darkolivegreen, antiquewhite, darkslategrey Description: a bedroom with bed desk and chair - Style: modern Palette: darkgray, black, darkolivegreen, darkolivegreen, dimgray Description: bedroom interior design ideas - Style: modern Palette: silver, black, saddlebrown, grey, dimgray Description: a bedroom with bed and desk - Style: vintage Palette: darkslategrey, lightgray, indianred, tan, grey Description: a bedroom with bed dresser and mirror - Style: modern Palette: darkkhaki, black, lightgray, saddlebrown, khaki Description: a bedroom with bed and book shelf - Style: modern Palette: lightgray, saddlebrown, sienna, grey, dimgray Description: a bedroom with bed and desk - Style: modern Palette: silver, darkslategrey, saddlebrown, grey, dimgray Description: a bedroom with bed television and - Style: modern Palette: darkslategrey, gainsboro, silver, darkgray, grey Description: a bedroom with bed and window - Style: vintage Palette: darkslategrey, silver, grey, darkgray, rosybrown Description: a room with bed chairs and table - Style: vintage Palette: silver, darkolivegreen, dimgray, dimgray, dimgray Description: a bedroom with bed and channel - Style: modern Palette: grey, lightgray, black, darkolivegreen, silver Description: bedroom interior design ideas - Style: modern Palette: darkgray, darkslategrey, sienna, gainsboro, darkslategrey Description: a bedroom with bed and sliding door - Style: modern Palette: silver, steelblue, peru, mediumaquamarine, saddlebrown Description: a bedroom with blue walls and wooden furniture - Style: modern Palette: grey, black, gainsboro, tan, darkolivegreen Description: a bedroom with bed desk and mirror - Style: modern Palette: silver, darkslategrey, sienna, saddlebrown, grey Description: a hotel room with bed couch and television - Style: vintage Palette: lightslategrey, darkslategrey, lavender, darkolivegreen, darkolivegreen Description: a bed with white comforter and headboard - Style: modern Palette: gainsboro, darkslategrey, grey, dimgray, grey Description: a bedroom with bed and mirror - Style: modern Palette: grey, darkslategrey, gainsboro, silver, tan Description: a bedroom with bed chair and mirror

Model-Generated Optimal Recommendation for ISTP Personality Type

Personality Traits:

- Preference for modern, sleek, and minimalist environments.
- Appreciation for neutral palettes with subtle warm accents.
- Value placed on simplicity, elegance, and practical functionality.
- Introverted tendencies reflected in muted tones and clean lines.

Design Recommendations:

- **Color Palette:** Neutral base with dark gray and black, accented by dark olive green and sienna.
- **Furniture:** Sleek, low-profile pieces in dark wood or metal with clean lines and subtle curves.
- **Textiles:** Tonal fabrics in gray, beige, taupe; natural fibers like wool and linen for warmth.
- **Lighting:** Minimalist LED fixtures and table lamps for calm ambiance.
- **Decor:** Sparse geometric patterns and modern art; clutter-free for serenity.

Room-Specific Suggestions:

- **Bedroom:**
  - Walls: Dark gray primary color.
  - Furniture: Low-profile bed with metallic frame and dark wood accents.
  - Textiles: Coordinated bedding and pillows in muted tones.
  - Lighting: Minimalist table lamp with LED bulb.

  - Decor: Modern geometric artwork with subtle textures.
- **Living Room:**
  - Walls: Dark olive green accent.
  - Furniture: Dark gray or black leather sofa and armchair, low-profile design.
  - Textiles: Neutral rugs and throws in beige and gray.
  - Lighting: Minimalist overhead fixtures with LED bulbs.
  - Decor: Modern geometric vases or metal sculptures.

5.16.2. Test Case 1: Evaluation Metrics and Results for ISTP

The semantic metrics—Semantic P-BLEU, Semantic ROUGE-L, and SBERT Similarity—are used to assess ISTP personality-aligned recommendations, as shown in Table 18. These metrics evaluate semantic alignment at phrase, sequence, and embedding levels between the generated recommendations and input prompts.

To provide an additional evaluation, the BERTScore metric—based on token-level embedding precision, recall, and F1 score—is reported in Table 19. However, due to its limited sensitivity to higher-level semantic structure and personality-specific alignment, BERTScore is not considered the primary comparative metric.

Observations: After fine-tuning, the ISTP results demonstrate consistent improvements across all evaluated semantic metrics. Semantic ROUGE-L increases from 0.375 to 0.828, indicating improved sequence-level coherence, while SBERT Similarity improves from 0.600 to 0.715, reflecting stronger embedding-level alignment. BERTScore also shows improvements, with precision rising from 0.763 to 0.803, recall from 0.808 to 0.810, and F1 score from 0.785 to 0.806.

Table 18
Semantic evaluation metrics for the ISTP personality type are compared before and after LoRA fine-tuning, showing a notable improvement in the resulting design recommendations' semantic alignment.

Metric	Before Fine-Tuning	After Fine-Tuning
Semantic P-BLEU	0.271	0.880
Semantic ROUGE-L	0.375	0.828
SBERT Similarity	0.600	0.715

Table 19
BERTScore evaluation before and after LoRA fine-tuning for the ISTP personality type. This metric is used for supplementary evaluation and not the primary comparative analysis due to its focus on token-level embedding overlap rather than deeper semantic and personality alignment.

Evaluation	Precision	Recall	F1
Before Fine-Tuning	0.763	0.808	0.785
After Fine-Tuning	0.803	0.810	0.806

5.16.3. Test Case Evaluation: ENFP Personality Type

Prompt Template for ENFP-Aligned Recommendation

Instruction: Create a personality-aligned home interior design suggestion for cluster "ENFP" by combining the specified personality traits and design elements. Make sure the design incorporates clear style preferences, color harmony, and psychological alignment.

Input: - Style: modern Palette: sienna, black, wheat, maroon, darkgray Description: a living room with couch chair and fireplace - Style: modern Palette: rosybrown, darkslategrey, antiquewhite, darkolivegreen, darkolivegreen Description: a living room with couch television and large window - Style: modern Palette: darkgray, black, saddlebrown, dimgray, sienna Description: a living room with piano and - Style: modern Palette: silver, darkslategrey, darkolivegreen, dimgray, grey Description: a staircase leading to the second floor - Style: modern Palette: darkgray, darkslategrey, dimgray, darkolivegreen, dimgray Description: a living room with couch and glass table - Style: vintage Palette: darkslategrey, tan, dimgray, darkgray, grey Description: a living room with couch and fireplace - Style: modern Palette: silver, darkslategrey, saddlebrown, sienna, grey Description: a living room with couch and kitchen - Style: modern Palette: lightgray, darkolivegreen, sienna, rosybrown, dimgray Description: a living room with couch television and flat screen - Style: modern Palette: grey, black, darkolivegreen, lavender, darkolivegreen Description: a living room with two chairs and table - Style: modern Palette: grey, darkolivegreen, darkslategrey, gainsboro, tan Description: a living room with ceiling fan and couch - Style: modern Palette: darkgray, sienna, darkslategrey, dimgray, dimgray Description: a living room with couch and television - Style: vintage Palette: grey, darkslategrey, lightgray, saddlebrown, silver Description: a living room with couches and chairs - Style: vintage Palette: darkslategrey, tan, grey, darkgray, darkgray Description: a living room with couch chair and coffee table - Style: modern Palette: grey, darkolivegreen, darkslateblue, gainsboro, darkolivegreen Description: a patio with blue furniture and ceiling fan - Style: modern Palette: dimgray, black, cadetblue, lightsteelblue, peru Description: a living room with couch and television - Style: modern Palette: dimgray, black, gainsboro, silver, darkgray Description: a living room with couch chairs and fireplace - Style: modern Palette: tan, black, sienna, saddlebrown, linen Description: a room with two chairs and table - Style: vintage Palette: darkgray, darkolivegreen, gainsboro, dimgray, dimgray Description: a living room with couch chairs and table - Style: modern Palette: peru, wheat, black, saddlebrown, silver Description: a living room with couches and fireplace

Model-Generated Optimal Recommendation for ENFP

Cluster ENFP Personality-Aligned Home Interior Design Recommendation
Based on the provided design features and personality traits, a cohesive and harmonious home interior design is recommended for Cluster ENFP.
Personality Traits:

- Modern and sleek design preferences.
- Appreciation for neutral palettes with earthy tone accents.
- Preference for natural materials and textures.
- Valuing simplicity and minimalism.
- Enthusiastic about integrating technology and innovative designs.

Color Harmony:

- Primary colors: Dark Gray (#333333), Dark Slate Grey (#2F4F4F), Dim Gray (#666666).
- Secondary colors: Sienna (#A0522D), Saddle Brown (#786C3B), Dark Olive Green (#556B2F).

These colors balance versatility with a natural aesthetic.

5.16.4. *Test Case 2: Evaluation Metrics and Results for ENFP*

To measure phrase, sequence, and embedding-level semantic congruence between generated suggestions and corresponding references, the evaluation used Semantic P-BLEU, Semantic ROUGE-L and SBERT Similarity semantic measures that were established in the INFP evaluation previously.

Table 20
An analysis of semantic evaluation measures of ENFP personality type pre and post-LoRA fine-tuning shows a significant increase in semantic consistency of design recommendation generated.

Metric	Before Fine-Tuning	After Fine-Tuning
Semantic P-BLEU	0.285	0.811
Semantic ROUGE-L	0.409	0.692
SBERT Similarity	0.638	0.811

Additionally, token-level embedding precision, recall, and F1 score were used to produce the **BERTScore** metric as an additional evaluation. However, as discussed in the INFP evaluation, BERTScore was not treated as the primary metric due to its limited sensitivity to deeper semantic coherence and personality-specific alignment.

Table 21
For the ENFP personality type, BERTScore evaluation is given both before and after LoRA fine-tuning, with an emphasis on token-level embedding overlap.

Evaluation	Precision	Recall	F1
Before Fine-Tuning	0.772	0.861	0.814
After Fine-Tuning	0.810	0.790	0.800

After fine-tuning, the findings consistently improve semantic and embedding-level alignment metrics, supporting the model's capacity to produce useful, personality-aligned interior design suggestions for the ENFP personality type.

5.17. Summary of Evaluation Results

The thorough examination of the personality type of INFP along with the further instances of evaluation of both ISTP and ENFP proves that LoRA fine-tuning significantly enhances the ability of the model to produce semantically consistent, psychologically based, and stylistically aligned interior design recommendations to be used in different personality types. With the INFP case, the model has reached almost perfect semantic alignment, which has represented the reflective and serene aesthetic feelings and interest in nature that are typical of this personality type.Also, the fine-tuned model has significant gains in Semantic P-BLEU, ROUGE-L, and embedding based similarity measures, tested in both ISTP and ENFP conditions, which shows the model is robust and can be generalized to diverse psychological and aesthetic settings. Despite the complementary nature of BERTScore, embedding-level semantic measures are more sensitive to alignment in personality-specific ways, which leads to the conclusion that the suggested framework is more focused on psychologically relevant and stylistically relevant design suggestions rather than focusing on the linguistic intricacy on the surface.

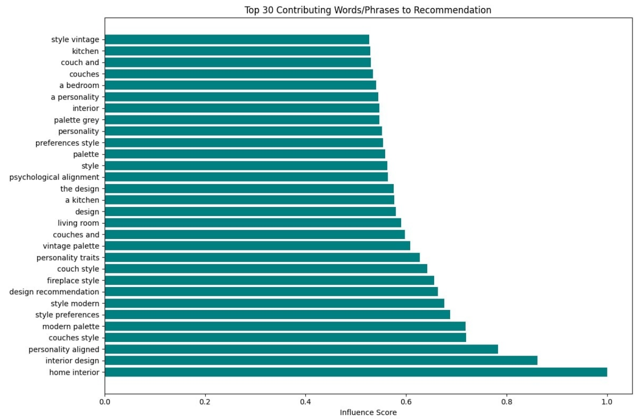


Figure 12: Normalized SBERT semantic similarity scores are used to generate recommendations, with key contributing words highlighted to show their influence on the final result.

5.18. Personality Prediction and Explainability of the Recommendation

We applied a two-tier explainability model to clarify and present personality-based interior-design specifications. The framework identifies two types of information (i) textual clues in the user input which has the strongest impact on the INFP personality classification, and (ii) linguistic components within the LoRA-generated recommendation which has the most significant effect on the final design. The differentiation of these two levels helps the framework to understand which aspects are influencing the personality inference and provide a systematic perspective of how specific words influence the recommendation.

5.18.1. Explainability for Optimal Recommendation Generation

We used the Sentence-BERT (SBERT) model to do a semantic contribution analysis in order to determine the most significant linguistic components in the provided recommendation. Using the **all-mpnet-base-v2** encoder, unigrams and bigrams from the user's descriptive input are included along with the whole LoRA-generated recommendation 5.14.1. Each input phrase embedding's cosine similarity to the suggestion embedding was calculated and normalized to a range of 0 to 1. The most significant contributors to the final recommendation were found to be phrases with the highest normalized semantic similarity. This method enables homeowners to determine which linguistic components most significantly influenced their customized design by offering a repeatable, comprehensible mapping from potential phrases to the provided advise.

5.18.2. Explainability in Predicting Personality

Word-level contributions for personality inference were obtained from SHAP values computed using a stacked BERT classifier in addition to recommendation explainability. These contributions provide interpretability for the system's personality prediction component by highlighting the words in the input that most strongly supported the INFP personality classification (Table 8).

Table 22
Unified explainability table for the INFP personality type showing the predicted personality, the referenced LoRA-based optimal recommendation, the top contributing words influencing recommendation generation, and the SHAP-based word contributions to personality prediction.

Predicted Personality	LoRA-Based Recommendation	Top Words Influencing Recommendation Generation	Top Words Influencing Personality Prediction
INFP	See Section 5.14.1 for the full optimal recommendation.	home interior (1.00) interior design (0.86) personality aligned (0.78) couches style (0.72) modern palette (0.72) style preferences (0.69) style modern (0.68) design recommendation (0.66) fireplace style (0.66) couch style (0.64) personality traits (0.63) vintage palette (0.61) couches and (0.60) living room (0.59) design (0.58) a kitchen (0.58) the design (0.58) psychological alignment (0.56) style (0.56) palette (0.56) preferences style (0.55) personality (0.55) palette grey (0.55)	peaceful (0.0360) feel (0.0360) cozy (0.0219) spaces (0.0219) imagination (0.0218) love (0.0213) where (0.0205) can (0.0205) practical (0.0195) prefer (0.0147) quiet (0.0119) am (0.0118) day (0.0118) colors (0.0093) enjoy (0.0093) soft (0.0093) explore (0.0093) cold (0.00829) loud (0.00723)

5.18.3. Graphical Explainability of Personality-Aligned Recommendations

In the suggested architecture, we came up with a graphics explainability module that will be used to display the system recommendations to the homeowners in a manner that can be understood and interpreted intuitively. In particular, the best interior design suggestions of the personality-conscious LoRA-fine-tuned model were converted into semantic embeddings and then divided into semantic contextual parts (in the form of rooms, furniture, colour schemes and ornaments). These embeddings were then prompted to a high-fidelity diffusion architecture, runwayml/stable-diffusion-v1-5, to generate photorealistic visualisations of

Figure 13: Personality-aware interior design recommendations are converted into semantic embeddings and visualized via a diffusion model.

interior design that corresponded to the predicted INFP personality traits and aesthetic preferences. This image-based elucidation approach allows house inhabitants to examine the spatial layouts possible, evaluate the congruity of suggested design traits with their own tastes, and have a concrete comprehension of the reasoning that the system used to make its choices.The module provides a viable and user-centred form of visual explainability in a personality-oriented content-based recommendation pipeline, as it enables obtaining prompt feedback, enhances user trust, and minimises the necessity of iterative redesigning of the process.

The proposed framework adds to the explainability of the contribution scores at the word-level, through associating them with the visually instantiated design constituents, and makes the interpretation of the explainability as a perceptually based interpretation as opposed to text-based justification.Semantic similarity scores and SHAP-based word-impact studies ensure that the personality extrapolation and the resulting recommendations can still be regarded as a response to the original input by the user, whereas the graphical explainability framework will translate those influences into real, spatial, and aesthetic features in a generated interior image in a house.Such conformity of textual attribution to visual display enables the users to look at first hand how certain personality cues and descriptive terminologies affect layout construction, choice of colour, and focus on decoration.

6. Summary of Results and Discussion

In conclusion, the experimental findings show that the suggested framework for personality-aligned interior design successfully combines interpretability, generative recommendation, and predictive modeling. Contextual embeddings obtained from BERT within a stacked ensemble architecture achieved high accuracy 90% in personality inference, while SHAP-based explanations provided transparent, word- and embedding-level interpretability. RAG-augmented Llama generation and LoRA fine-tuning considerably improved the subtle, customized recommendations made possible by the mapping of MBTI types to interior design preferences. After fine-tuning, evaluations reveal notable improvements in the stylistic and semantic alignment of the provided recommendations. INFP personality type recommendations received BERTScore F1 of 0.81, P-BLEU of 1.0, ROUGE of 1.0, and SBERT similarity of 0.8, demonstrating the model's ability to capture both the aesthetic and psychological aspects of interior design. Other types of MBTI, including ISTP and ENFP, have been tested on additional test cases, which further confirm the generalizability of the framework. ISTP-aligned propositions focused on practical layout, plain furniture, and neutral-to-cool colors but ENFP-aligned propositions involved bright accent colors, numerous textures, and inventive use of space. The Semantic contribution analysis was used to determine the key language features which informed the creation of type-specific recommendations in both examples and SHAP explanations ensured the fact that the model properly equated the identified textual indicators to the predicted personality type. These findings confirm the effectiveness, interpretability, and usefulness of the framework in providing personalized and personality-based interior design solutions to diverse MBTI profiles.

7. Conclusion and Limitations

This research work proposed a novel framework in giving personality-driven interior design recommendations that align with the intrinsic preferences of the homeowners. The proposed methodology does not rely on explicit user inputs like sketches, purchase histories, or personal data, which are commonly employed in design suggestion systems. Instead, it predicts the personality type of the homeowner by considering textual descriptions and generates recommendations based on aesthetic and psychological considerations. As a result of this approach, the manual effort required from designers is reduced, while it enhances the satisfaction of the homeowners by minimizing multiple design iterations. The proposed method accurately infers the MBTI personality types by effectively stacking an ensemble of machine learning models with deep natural language processing methods like TF-IDF, GloVe, and BERT embeddings. The system provides explainability through the use of SHAP,allowing users to identify which terms influence the expected personality and the generated recommendations. Following the mapping of interior design elements to each personality type using CLIP and BLIP embeddings, a retrieval-augmented generation module ensures that contextually relevant design recommendations are provided. LoRA fine-tuning further refines the model for semantic, stylistic, and psychological alignment in the given recommendations.After fine-tuning, evaluation demonstrates significant performance gains. P-BLEU rose by 28.9%, ROUGE-L by 70.3%, and SBERT similarity by 36.3% for the INFP profile. P-BLEU improved by 60–65% and ROUGE-L by 55–65% in test cases for ISTP and ENFP. These findings confirm that the framework generates suggestions that are psychologically consistent with the homeowner's personality in addition to being semantically and stylistically correct. Additionally, the explainability layer improves usability and trust by offering transparent, traceable insights into the rationale underlying specific recommendations.

Although these results are promising, this research study is limited in a number of ways. The lack of a publicly available dataset with mapping personality traits to interior design preferences is the primary limitation that limits the variety of training data and the generalizability of the model evaluation. In addition, other data modalities,such as. the images of the existing spaces, verbal descriptions and room dimensions are not included in the current framework, which is restricted to the textual input. Despite its thoroughness, the evaluation measures used BLEU, ROUGE, SBERT mainly assess stylistic and semantic consistency, but do not reflect personal satisfaction of homeowners and experience. Lastly, the two-dimensional, picture-based presentation of the derived recommendations may not be sufficient to convey the spatial depth or refined visual qualities, thereby forming another limitation in fully capturing the intended realism and perceptual richness of the generated outputs.

CRediT authorship contribution statement

Zainab Asif Jawed: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Resources, Software, Writing – original draft, Visualization. **Muhammad Rafi**: Methodology, Software, Validation, Investigation, Writing – review & editing. **Syed Zain-Ul-Hassan**: Supervision, Project administration,Data curation, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are not publicly available. The experimental setup combines the MBTI personality dataset with a collection of unlabeled interior design images. As no benchmark dataset exists that directly links personality types with interior design attributes, we construct an intermediate representation by extracting visual features from interior design images and transforming them into structured textual descriptors. These textual representations are then used for mapping with personality traits in the proposed framework.

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